

Geo-spatial modelling of vehicle crime in Northern Ireland using computer vision to identify environmental factors.

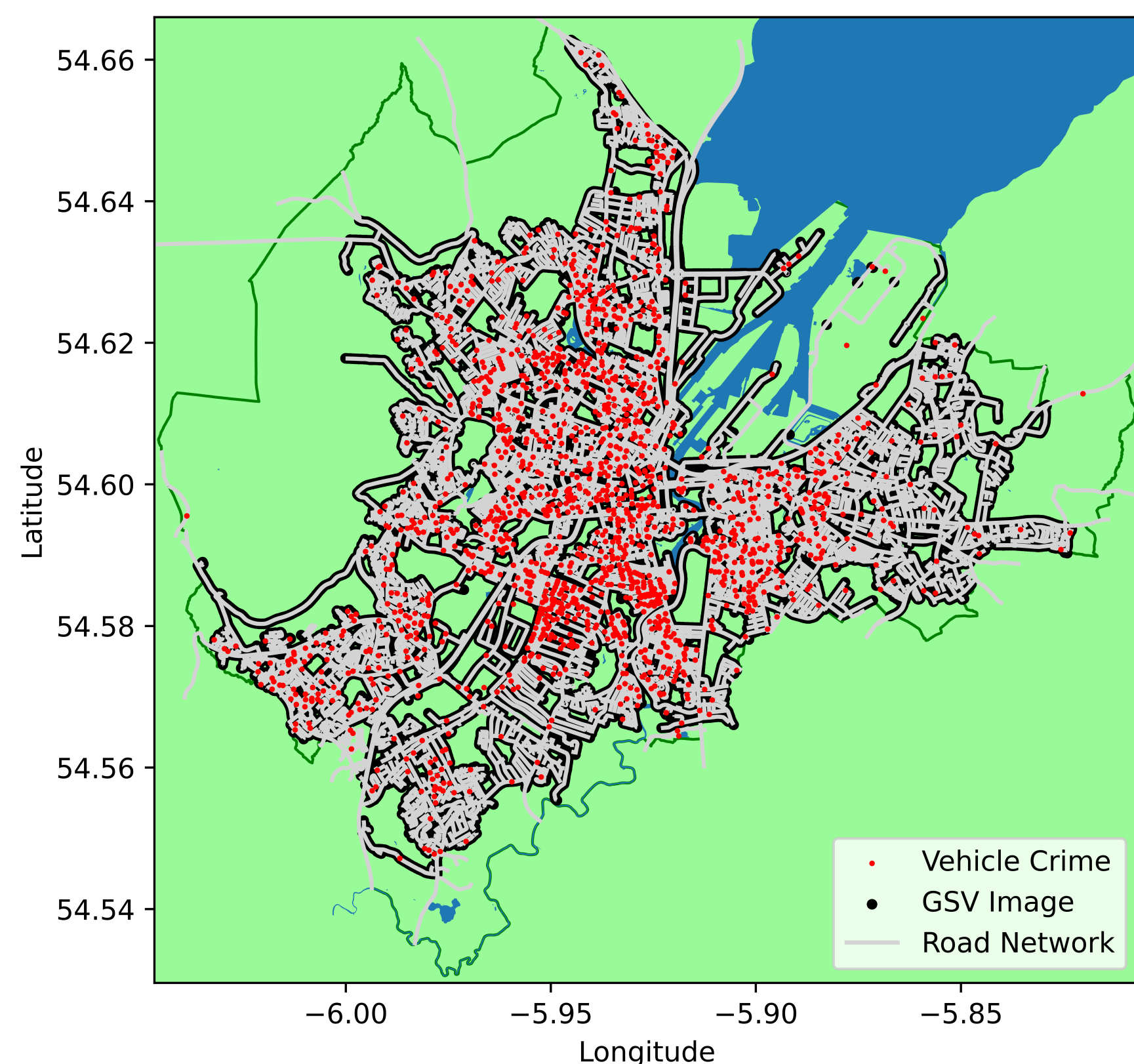
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Introduction

Our research uses computer vision (GroundingDINO[1]) to identify environmental factors which may enable or deter crimes, particularly vehicle crime, from occurring. We build associations between detected objects in Google Street View (GSV) images and vehicle crime through regressions based on the prevalence of each object in a location. We propose that this methodology can be used in actuarial applications as a superior alternative to postal codes.

Computer Vision



(a) Map of Belfast City showing the road network, location of GSV images and Vehicle Crimes.



(a) Example of GroundingDINO correctly identifying cars. Image: Joanmount Park, Belfast (01/08/2022).

References

- [1] Liu, S. et al. [2023], 'Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection', *arXiv preprint arXiv:2303.05499*.

Data

GSV Images

- We have access to over 1.9 million GSV images for Northern Ireland, taken between July 2008 and July 2023. Our primary focus has been on Belfast City. GSV images were taken in 55,415 locations within this area, with 6 images per location to create the 360 degree view. This totals 332,490 images for Belfast City.
- The images are recent with 84% sourced between 2020 and 2023, and are mainly taken during the Spring and Summer time, with 94% being taken between March and August.

Vehicle Crime Data

- We sourced publicly available vehicle crime data from <https://data.police.uk> to overlap with the majority of the GSV images for Northern Ireland.
- Between September 2020 and July 2023, 2,700 vehicle crimes took place in Belfast City.

GroundingDINO

- GroundingDINO is an open-set object detector that combines DINO (a transformer-based detector) with grounded pre-training.

There exists a wealth of literature hypothesizing that criminality is influenced by environmental factors. We look for signs of decay and dilapidation in GSV images by detecting broken windows, abandoned buildings, vacant sites, overgrown vegetation, graffiti and litter. We also look for deterrents to crime such as the presence of police, amenities, self-policing communities (through paramilitary flags and murals) and surveillance devices.

Cars	PP	PN	F1 Score	Precision	Recall	RMSE	MAE
P	1383	179	0.8879	0.8905	0.8854	1.2145	0.5817
N	170	73	-	-	-	-	-

Table 1: Confusion matrix and resulting metrics from GroundingDINO model detecting cars in a sample of 600 randomly sampled images in Belfast City.

Actuarial Application

- Insurers use postal codes to assess the level of risk associated with an area by pinpointing the location of losses, e.g. theft. This assumes historical data is a suitable indicator for future claims, which may be false if changes to the environment occur.
- We propose that by finding indicators of the presence or absence of crime in GSV images, we can address why certain environments have greater levels of risk. As the environment changes, we can better account for the fluctuating risk levels in areas.

We model the number of vehicle crimes in a location using a Poisson regression: The model takes the form,

$$\log(\mathbb{E}[Y|x]) = X\beta$$

where Y represents the number of vehicle crimes, X is a matrix of regressors and β is a vector of coefficients. The final model found the number of vehicle crimes in a location increases with the presence of litter, shops, graffiti and cars, but decreases with the presence of street lights.

Car	Graffiti	Light	Litter	Shops
1.0698	1.2623	0.6745	1.4479	1.2860

Table 2: The factor the number of vehicle crimes changes by for a unit increase in the respective regressors.

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